

ActInf GuestStream 120.1 ~ Temporal Cognition and the Emergent Properties of Consciousness

GuestStream | Temporal Cognition, the Emergent Properties of Consciousness | 1:06:31

Hello, welcome everyone. It is September 17th, 2025. We are in active inference gueststream 120.1 with Emanuel Adfa Olasupo who will be presenting modeling thought in time, a physics of temporal intelligence as the next frontier of AI. There will be a presentation followed by a discussion. So if you're watching live, please write any ideas or questions in the live chat and thank you for joining Emanuel to you for the presentation.

>> Thank you Daniel. I appreciate the um the invitation. Um it's uh I'm incredibly humbled by um you know the opportunity to to give this talk. uh the title actually is not model in time. uh but this is the paper that I think uh connected you and I uh and so I just wanted to start the presentation by sort of uh clarifying or motivating the talk based upon this paper this preprint that I had published um and just to give a short summary of what the paper is about um you know the argument is that the uh current LLMs that um are being used right now both in the industry and in academia. They um have what we call a temporal blindness problem which means that they um do not represent time um in the lived sense right as as humans do. And so if the goal is to achieve conscious AI or AI that mimics human cognition um and humans ability to manipulate thoughts and emotions then uh

this paper argues that time is um a foundational uh feature or uh that needs to be represented akin to how humans um experience time in the sense of thinking in time and so on. I'll get back to this towards the latter part of the presentation but I just wanted to motivate

uh the the lecture uh based upon uh this paper the the the actual title of the talk is uh uh temporal intelligence and the emerging properties of consciousness um indeed you know I'm going to sort of to break this talk into three components the first is I'm going to idea of course defined and process it, right? And so this is a type of process in that um one would argue supports temporal cognition by structuring how information is organized over time and enabling the system to become more predictive and generative and generate expectations about the temporal sequence of impendent inputs. Right? And so that coarse course defined process and that's the first part that I'll talk about and I and I'll talk about the mathematics that perhaps governs that process both within biological systems but also artificial systems.

The second part of of this of this talk is going to be uh on this idea of predictive remapping. Um and and this is an example of what I would think of as temporal cognition and action, right? Um um this is informed by um this idea of active sensing, right? Uh this idea that um cognitive systems must move in order

to compute information in the world,
right? And so I will also uh speak a
little bit about this. Um and then the
third part is the more um um
speculative or perhaps more um has is it
has the potential to be more uh paradigm
shifted if um one or myself or groups or
others anyone can achieve this this idea
of uh the emergence of thought right so
so how does your thought emerge how do
they flow and synchronize with other
people's thoughts and Can we achieve a
mechanistic view of consciousness?
Right? Because that I think is key into
achieving AGI, right? Artificial general
intelligence. And so th those three will
be the so so this typeight structure
would be kind of how I um uh go about
giving this talk.

And so
you know I normally approach questions
with physics and mathematics. All right?
So I'm a theoretical physicist. My my my
case is theoretical physics. But you
know I I I normally use the two to try
to understand the phenomenons that I
study and I particularly look at complex
systems particularly biological systems
to explore that and so that is important
to sort of highlight my biases right we
all have biases in which we use to sort
of inform how we think about the world
right um
so this idea of course defined I've I'm
illustrated here in this figure right So
what you see here is at least hopefully
what you can appreciate here is that as
you move towards the right this scene

gets more fine right and this is the general principle of course defined processing and so what I'm showing here is a cartoon of neurons in the brain um that turns out they compute each part of this processing right and so you have these these early cells this this where I've represented here late cell bifphasic and late and so it turns out that in the in the in in the mice's brain there are neurons that compute each part of this kind of uh uh stage in course define and you know you can sort of see in this cartoon here or so in in this in in this experiment with Parker at all where you have the mouse moving along this vert virtual world right and its neural activity is being recorded during that sort of task that is being asked to perform and you know the work this work here that I'm presenting is done with Christopher Neil at the University of Oregon and he basically pioneered the field of active vision and what's very profound about this is that this is such a break from how bi biologists and neuroscientists do experiments because now these animals aren't headfixed right? But they're actually moving in the world akin to how we as organisms move into the world and collect information. And so in this paper they found this coar defined processing phenomenon which I've illustrated again here in cartoon right but I I want to emphasize that coarse defined processing

is quite um ubiquitous and what that actually means is that you can actually um you know there are papers that have published course defined processing in other sensory modalities herin smell taste touch and so let Allow me to give you a simple example. You can imagine you are in a group of crowd and you meet your partner for example and you you spot her at a distal location. What you might first do is sort of hear this sort of global and coarser voice of everyone and then as she approaches you closer you um kind of finetune the your auditory uh uh system in a sense. your brain will do that and then you start to focus on a singular voice. And so that's an example of this course defined process and operating in the in the auditory system, right? And so you can find this process um across sensory modalities and it's not just across biological systems but it's also artificial systems, right? And so the lost paper that I cited here, you know, they use coarse defined process and to sort of optimize the grasp or the grip sorry of of robots. So you can imagine a robot extends its arm, tries to reach for something. It first needs to kind of get an understanding of the general area of the scene and then over time we'll understand oh this is a cup so I have to hold it in a particular way in order to maintain the grip right and so course define excuse me the point that I want to make here is that it's a it's a ubiquitous way by which biological

systems and artificial systems process information right and so what I am interested in understanding is what are the mathematics what are the computations that these systems are performing in order to achieve the output that I've shown in this sort of nice illustration, right? And so the first thing, you know, um and so and so let me dive into that a little bit. So and and and again I I I am aware that perhaps people that are watching have different um um come from different fields. So, I do apologize if if it's a little bit too complicated or or um I haven't explained it um as as as as as well as I would have liked, but let me take a stab at this. So, what you see here is we're going to we're going to try to understand this phenomenon using what we call a recurrent artificial recurrent neural network, right? And so the hallmark of these networks is one they are connected right to each other. So let's just say for example you see E and I there's a line that connects E and to I and I to E. They also have self connections, right? So you see that E is connected to itself. I is connected to itself and they also have the ability to to to to to get input, right? External input. And so these features or or or these components make up a recurrent neural network. And they also go in multiple directions, right? So it's not one way, right? like a feed forward direction but also

feedback and so on and so on. And so people in the past have used these networks to try to understand phenomena in the brain for example. And one of those is Murphy Miller uh 2009 where they published this paper called balance amplification.

And so allow me to try to explain a little bit of what this means. What does it actually say? Right. So here is so we're going to be working with ENI networks right and these are excitatory and inhibitory networks in the brain in the biological system and so here shows the equation we call this a rate model equation first differential equations where you have E here which represents the excitatory parts of that network right and then you have the inhibitory parts of that network so we have R which represent the rate right of the network we have W represent the weight matrix

And then you have I which represents the um internal or external input that the system will get. And so you have this first order differential equation that basically will tell us we think of this as the governing equation. It will tell us how the network behaves over time.

And so this is the tradition that I was working from when I wanted to understand the mathematics governing uh course define. And so what I have done is basically I took the same governing equations right. So E here just

represents I the same way. So this represents just the external input and I have solved this equation in a fundamentally different way. And so that equation and the solution for that equation is what we call what we think of as uh the course defined solution. And so this solution here, what is unique about it is that instead of each unit in the network having a single time by which it it's perturbed or responsive, it actually has a heterogeneous uh uh um it has the ability to have each neuron has the ability to be to have its own temporal dynamics. And you can imagine if you want to understand something like coarse define, you need a temporal handle in order to kind of play with and parametrize in order to try to simulate whether or not the model can achieve the particular output. And so that is the the the the novelty of solving this particular equation in the way that I've solved it. And this is kind of like the first type of this is the this is the first time that such a equation has been solved in this manner. And I think of this as a general form of this uh solution because it can be generalized to a lot of other things, right? So you can imagine it could be generalized to um you know um uh not just vision but you know different sensory organisms uh sorry different sensory modalities like audition and so on. as so long as it has this handle of time and something that

is trying to um uh uh um some a preferential uh object in this case let's say vision right and so um this um cartoon if you may right this is the case when if you were to draw a diagram of the neuron um two neurons for example right so E and I but this diagram is basically showing more of a population representation of the network and we're going to think of the network as not just E and I but we're going to think of them as having two different modes. So you can imagine the network interacts with each other and E subtract and you have what we call the difference mode. So you have E and I whereby they're kind of competing with each other, right? And you have another mode where you have E plus I and the sum and and they're summoned together in a sense, right? to produce some type of effect. I know that's a mouthful that's a little bit hard to process, right? But I just what I want you to take away from it is that when we look at these networks, we're going to look at the modes by which these networks operate within. So we have the sum mode and we have the difference mode. And it turns out that the difference mode, right, is what actually creates the initial amplification in the system. So let's say for example, when E and I are imbalanced, you have this initial amplification. That's where you can that's how you can maybe understand the initial parts of this of this time

series. So in this case here E and I they're imbalanced the network amplifies right based upon its feed forward connection to E plus I and then as this minus W plus uh gets stronger you get an inhibition and then the network goes down.

And so this is the type of network that and so this is the type of behavior that we would expect.

And so remember the goal here is to simulate um coarse define processing right and so on the on B here is the experimental results all right that that I've shown um earlier in in in the cartoon right and then here is the the the response of the network that I just showed you when we don't solve the equation the way that we uh the way that I've solved it uh with the course defined solution and you know you don't have to be genius to realize and see that you know these two are not the same. Right? And so what this is saying is that the way the the the way by which these governing the Murphy Miller solved this equation cannot produce coarse define.

But what if we used the coarse defined solution the one that I noted earlier?

What you start to see is something a little bit it starts to resemble B , right? But here's the catch.

This is this simulation here is the case where there is no input into the network. So this is actually imagine you have a netbook or you have a system imagine I took a ball and I put it on

top of a hill right because of the and
and I didn't touch the ball at all
because of the uh the structure or the
architecture of of the hill the ball
will just roll down right so you can
think of that as the internal dynamics
of the system where there is no no one
pushed the ball there it's just
basically the system that you're trying
to understand and how it behaves and So,
but at the same time, you see that it
doesn't quite mimic B. Now, if I create
an input into E and I,
you can see that D rep D and B looks
very similar. So, what does it actually
mean? What does this actually mean? What
this means is that when you think about
course define, when you think about
systems in general, it seems that
systems have this internal structure
that is almost tuned or prepared to
behave or be amplified in a way that
represents the way in which the inputs
um kind of favor if you may, right? And
so basically what this is saying is that
you know in short
um you can understand course defined
using these governing uh this solution
that I that I have um that I have um um
um sort of put forth
e just to not complicate things any
further
this or this is my I can't get feedback
from anyone but Danny can you see my
screen can you see my m Perfect. So here
you see that E so so so on the Y axis
here is the kind of distribution of how
I set the time constants of each neuron

because if you remember in in my governing equation each neuron can have a different time constant right and so this here shows that for this kind of like let's take for example this trace here that represents a certain class of neurons neurons, the early neurons. I set the excitatory neurons to be slower than the inhibitory neurons, right? Because you know the higher the value, the slower, the lower the value, the faster, right? And then you can see that for this set of neurons, I set it to be the same thing, right? Excitatory neurons are faster than the inhibitory neurons, but you can see that the net average of it is slower. So you see this difference between excitatory and inhibitory. You see the peaks different. But you can see for example the biphasic it's almost a flip where the excitatory neurons are slower or faster than the inhibitory neurons. And so this is something that now the experimentalist can go and say okay let's measure specifically the inhibitory neurons and see whether or not they actually represent or recapitulate the model's predictions. Right? And this is kind of the point of like why you know some you know for those that are not necessarily uh interested in modeling and mathematics you might wonder like so why why model these things right it's because these models will generate hypotheses and intuitions that

experimentalists can now go and test and say ah okay you know uh maybe in the bifphasic neurons right in the bifphasic group the one that looks like an S right based upon Emanuel's uh uh uh uh uh predictions

We expect the when we measure the time of the inhibitory neurons, we expect them to be slower than the excitatory neurons, right? And so this is kind of um you know what you know we did when we tried to understand uh whether or not these equations actually recapitulate the biology.

So you know so that's the first part of what I want to talk about this idea of course define and how it kind of it kind of gives you a hint as to how systems kind of are how how the functional architecture or the or the structure the internal dynamics of the system is almost prepared to process information in a way that makes sense right in a way by which is economically efficient or energetically efficient right and so the second part I'm going to talk about this idea of what I call retinal topic mechanics.

This is a little bit more um of the physics uh side of me informing um um some of these uh questions that people have been trying to solve for for many years. So um should I wait for questions or should I continue to go Daniel because it's it's a bit of a first part. Let's continue.

Excellent. Okay. So the the second part

here I'm going to talk about this idea of predictive remapping. And so retinal topics is remapping is basic retinal topic mechanics is basically a field that I have put forth right a hybrid of physics and neuroscience.

And you know the the the the the the framework came about based upon this notion of or this idea of um uh uh um um uh um um motion blur that happens when you make eye movements. So here I'm showing a human eye. Here I'm showing the mouse uh the the the eye of a rodent making eye movements. And then here I'm showing the eye of a m of a mount um sorry a drosophila which is like a a fly in a sense right and you can see that all of these systems across scale they they make eye movements right they they actively make eye movements around the world to try to collect information right

and one of the things that that happens is that whenever you make an eye movement

you experience this kind of motion blur right like but sorry you don't experience it but That's actually what happens. And so your brain needs to find a way to kind of eliminate and cancel this motion blur so you can see and experience the world in a very continuous and fluid manner. And so one of the questions that neuroscience and physicists and mathematicians have been trying to answer for many years is how does this happen? Why is it that you know you don't experience what we you

would experience if you were like taking a picture with a camera for example.

Imagine if the train paused you and you try to take the picture right of this

fast train with the camera. You would experience the blurriness of that train.

But you and I make these eye movements all the time. If you think of the eye as a camera, but we don't experience this motion blur. How does the brain achieve that? Right? That's really the question that informed this uh this this computational framework that I introduced.

And so many people think of neurons as perceptrons, right? or like a thermometer, right? Where you know, imagine you had a thermometer, right? It would give you the measurement and it would go back down, right? Or perceptron where it's a linear weighted sum of the inputs that you put into the network.

And that's basically the foundation of, you know, AGIS, right? Where you have these perceptrons and and they're performing these operations.

And so with this, you know, I radically turn things on its head and I think of these I think of neurons as spring-loaded senses, especially because I'm interested in this idea of movement and active sensing. So it's not that the neurons themselves are moving, but they're they're they're they're getting information from different parts of the brain that is almost um tethered to where in space they need to acquire information. Right? So you can imagine

if I wanted to look here, right? Because I have these two monitors, so it looks like I'm looking away. But imagine if I if I wanted to look here, right? Right. I would have to sort of allocate some resource towards this part of the visual scene, right? In order to achieve some level of uh perception, right, or some desirable outcome. And so what you see here is not the neurons themselves moving but how the brain or how these neurons are operating like springs in order to allocate different uh neural uh resources in different parts of of space in this case uh visual space. So A here is sort of um you know uh this spring sort of being attracted to this moss right um b here is a case where it's unresponsive. So you can imagine a maladaptive case where you know you don't want to pay attention to something and so you you self-inhibit right or maybe a disease model where you can't pay attention ADHD you can imagine hair C being a case where you hallucinate right so instead of you responding to N you start responding to something else right um and this larger uh uh uh field that you see is what I refer to as elastic fields these things have no uh empirical support yet, but there is something that I think is foundational in understanding active sensing and and active vision, right?

And so I'm not going to go too much into details of this model. There's a paper that you can read uh read where I published in physics review where I

outline comprehensively this these equations and and how the prediction goes. But in short, I bring together a range of papers that show that hey, like maybe what the brain is doing it's it's it's it's undergoing these predictive shifts prior to an eye movement. And these predictive shifts are what kind of blunts

uh this motion blur that that would normally occur. Um but what I found in this paper just to just to conclude is that these uh these uh these uh these processes right these predictive processes they're all happening but they're happening at distinct windows in time. So for example you can imagine if I wanted to look at an object that is here

the peripheral moss would be the thing that is actually pulling the system prior to me making the eye movement. And so that predictive remapping of the neurons right of the springs that would go prior to me making an eye movement such that I wouldn't actually see the intermediate parts. So it's almost like the it's almost like the resources that I have here would kind of predictively go here so that when I make the eye movement there is no resource that is actually computing the intermediary parts for me to actually experience the motion blur. And so that's the general gist of this kind of idea of predictive remapping. And my contribution is that I think that the computations and the and and the and the way to think

about this is in is in the terms of
forces is in the terms of Newtonian
mechanics. And uh like I said um you can
look at the paper um um if if you want
to know more ask me questions after uh
this talk. Now let's get to the more fun
part of this talk rather than the more u
I guess uh

technically intense part. So ultimately
right like you know the connection is to
understand these temporal dynamics in
the brain in order to be able to
understand consciousness and eventually
apply it to artificial intelligence
systems. And so the first part of this
is to understand

um consciousness, right? And you know I
think that you know I normally use this
example as a um as a um
as a metaphor to understand conscious or
conscious or conscious behavior or
collective intelligence.

So what you see is that you know this
process called murmmoration by the way
and so what you see is that each of
these birds right they are on one level
individuals

but on another level they're a
collective they're part of a group
they're part of a larger network
and what you see is that you know at
some level of description you know you
see it looks like a chaotic system very
random very stochastic but then at
another level you start to see this kind
of collective behavior, this emergent
property where they're kind of moving in
a very structured manner. And so I

normally think about humans in this sense, right? I normally think of thought in this sense or even neurons. And I'm not the first to think about this. Giorgio Perezzi who won the Nobel Prize, a physicist kind of used the bird flocking as one of the uh systems that he studied to understand complex systems.

And so one of the things that I'm doing right now is to understand um these systems in terms of dynamical systems in terms of attractors specifically and you know many people have done this before but not applying it to the question or to the phenomena that I will be applying to which I'll talk about in the next slides and so we have these things called attractors in physics and you know here I'm showing you a strange attractor a tourist attractor a cycled attractor and then here's the point attractive.

So just to give some intuition, right? You can think about a pendulum. If I had a pendulum for example and I move that pendulum and I let it go, right? That pendulum at some point will stop right at the center. So there's a particular point that's attracting that pendulum, right? And that's an intuition that can give you about what a point attractor is, right? A simple pendulum, right? the the dynamics of that of that of that system reflects a point attractor.

And so is there a tractor? Is there an attractor that can help us understand

consciousness? And specifically what I mean in this term by conscious is just mental thought, right? I think of mental thought as or mental content as kind of something you need to understand in order to understand uh consciousness or put forth a theory of consciousness. And so one question that I think needs to be asked is or or or or something that I believe is that to understand consciousness and leverage the insights to build conscious AI, we need a database to use to study and ultimately uncover the fundamental laws governing consciousness.

And so I would argue specifically in the scientific domain in the academic there is no such database.

Right? at the very most you have consciousness researchers looking at maybe 50 subjects or two people in the lab and an fMRI study and you know it's I don't think that data is sufficient in order to understand something as dramatic if you may as consciousness and here's an example I want you to ask yourself like is there someone in the world right now having the same exact thought as you at this very moment in time right and if there was How would you even measure this? Right? There is I mean I'm sure Google has this data but they're not interested in actually studying consciousness. They're interested in selling you that you don't want to buy. Right? But in short, right? Like from you know when you think about the study of consciousness and the

history of consciousness there's not a lot of there isn't any data set that you can actually mind or study to explore to actually give you intuitions about what possibly could be consciousness or collective intelligence and how you could even apply that to an a conscious AI right if if if such was even possible.

And so that is what led to the this you know the the introduction of conversation net. And so conversation that is a repository that um has spontaneous human conversations in real time that you can actually use as a way to measure whether or not uh for example is there someone saying the same exact thing as you in this exact time. Right? You can start to understand these types of questions or at least study these types of questions.

And so we just released uh the third uh database for conversation net. And just to give you an example of that, you can sort of go on the conversation net uh uh uh uh uh um website and you can just download this data for free and uh sort of um you know um sort of try to understand or try to sort of mine this data, right? And a lot of this data are anom anonymized uh consent of people and they're coming from my other company Figbox that uh you know one of the core attributes is connecting people uh through uh uh uh 20-minute conversations. And so I just found that

as a as a as a as a as a as a useful way to
uh sort of u um support scientific
research. And so you know there's there
are some people in the audience that may
be aware of imageet
which is a database that basically uh
was foundational to um uh introducing
data into machine learning systems and
as a result you know ImageNet was used
to build AlexNet uh and u you know uh
here's professor Hilton who recently won
the Nobel Prize last year but and this
is his student uh Alex who built AlexNet
and So a lot of this uh we think that
you know conversation that is is is is
really important into uh not just
studying consciousness and complexity
but also uh can help in building
conscious AI more humanlike AI right
studying human conversations spontaneous
conversations and so on and you know
Alex net led to let's say for example
some of the software is used for facial
recognition self-driving cars and so
forth
and so what are some of the things that
we want to use on conversation as far as
like to study uh uh uh um these types of
uh uh properties, right? One of the
things we you know some of the things
can be this idea of self-recurrence,
inter aent synchronicity, you know,
phase relationship and variance and then
uh propagation and variance. And so let
me just give you a simple example of
that. So here's a case of of of
someone's conversation, right? This
individual was having a conversation,

right? And what we did is we measured
I'm sorry. I'm showing an instance where
the person is saying the same exact
phrase at different points in time. And
what you can see is you can see kind of
this structure in this uh in this uh uh
in this uh what we call mocognitive maps
where you kind of see the same structure
emerge when the person is saying the
same thing at the same ex um at
different points in time. And so this is
the some of the ways in which you can
start to measure
um and and and visualize right these
sort of things that I've uh these sort
of u um these sort of phenomenons that
we're interested in exploring and we
think of these phenomenons as temporal
invariance right things that kind of
have this predictive pattern that you
can kind of tease out from this data set
in order to understand thing uh uh the
idea of u you know whether or not you
know um there is some emergent property
happening as people are having different
thoughts in different speeches in
different parts in space.
And so here is just a simple example of
how you would think about um um um
attractors, right? So what you can see
is that there are four attractors in
this kind of fictitious uh uh
simulation. And each individual, you can
think of that as every individual having
some thought, right, in some time
window. And so each individual in some
ways are being attracted to a particular
type of thought. And so let's say for

example some famous person just died
right um you know you can imagine that
that you know a lot of people are
talking about those things and so
they're attracted to that thought right
within some window in time right and so
this is kind of again this is fictitious
right this is not actual data but this
just gives you an idea of how you can um
how I'm thinking about using attractors
to understand and and and uh uh uh this
idea of of thoughts or consciousness and
again how you can start to mind this
type of data to to ask these very kind
of sci-fi questions that you know um
prior to conversation that you you
really can't answer these questions
because there's no data to answer it
now you know ultimately right you know a
lot of what I just talked about is very
new and you know very althogonal and
very different from the way people think
about these things and I understand that
um and So what we can just do is, you
know, is there a way for us to learn
something about the data that we're
we're we're measuring and just add some
sort of temporal invariance layer to
these networks that, you know, OpenAI
and Anthropic and Gemini, right? These
LLMs, right? These uh these uh
transformers, right? And you know, I'm
not particularly um um uh optimistic
about that path. I think to get AGI we
need to fundamentally build new systems
that are radically different right but
one of the things that we've done um is
build digital twins

right and so this is just a this is and
this was done actually a lot I I I've
included these individuals that you see
at the bottom because they kind of
inform the way that I think about they
kind of teach me to be honest the way
that I think about uh artificial
networks so this is um someone like this
a brother to me uh Kuramusa
uh you know he's a machine learning
expert in in London and then this is
Daniel Olds he him and him him and I
work uh with uh the conversation at a
database and we try to understand
systems and uh I should mention that
this right here is actually uh uh for
the most part built and trained by uh
Timothy um who um uh is another uh young
brilliant individual that um uh works
with us on conversation. And so
basically what we did is we we we took
these LLMs and we trained them on
conversation net data to see whether or
not they improve performance or not. And
so in short, you know, there are ways in
which we can use the LLMs now to try to
understand whether or not uh they will
sort of create more humanlike responses
and things like that. But like I said,
I'm not particularly convinced. And I
think um the endeavor of maybe uh
building new types of systems uh could
potentially be fruitful and
groundbreaking. And so I go back to this
paper that I initially uh brought up in
um uh at the start of my talk um where
it's like how do we build systems that
think in time? How do we build AGI that

think in time are predictive that
can maybe create causal temporal
inferences and things like that.

So,

I've been sort of talking to a very um
you know, you know, he he's he's pretty
out there this this guy and and I love
him for that. And him and I talk a lot
about systems and um how to think about
systems and um you know, uh Rob, you
know, we've been thinking about and and
sort of playing around with this idea of
polychronous systems. So, let me explain
that a little bit.

What I'm showing you here is a network
that has two layers. Layers I and layer
J.

P I P2 and P3. They represent uh let's
say C A and T.

What defines a polychronous group is the
temporal pattern

um that um that leads it to let's say
for example the second wave. So in this
case here P1 is connected to cat with a
latency of one. P2 is connected to cat
with a latency of five. P3 is connected
to cat with a latency of nine. Right?

For this word cat to be encoded
there needs to be an activation key and
a delay key. So in a sense the
activation key needs to be uh configured
in a way by which they all activate this
node at the same exact time. And so you
need an activation key that is almost an
offset to the delay key. And so if you
were to add this together like let's say
for example we add 8 to um 8 to 9 8 to 1
sorry you would get 9 0 to 9 you would

get 9 four to 5 you would get nine. And so in a sense if you just took this very simple example as the initial measure or the initial way by which the network is computed or encoding words. You can kind of instantly see that it has to do some temporal computation for it to encode a word. This is radically different from let's say the way LLMs are now where you have let's say C as a vector that has a series of numbers. You have A as a vector with a series of numbers. You have T as a vector with a series of numbers. And they're just basically doing these statistical correlations with these tokens. Right? But what's happening here with these polychronous systems is that the system is actually having to think about the temporal order and the temporal differences in order to encode time, right? And so sorry, in order to encode words and so what you can get is an increase in representational capacity, right? You can also get emergent behaviors like self-organizing behaviors. I don't want to go too much into that, but I can I'm more than happy to explain what that is in the in the question section. But in short, right, like this is kind of the the the work that I've been doing uh from a both, you know, from from from the biological side and the stuff that I've been doing from an artificial side. Again, I understand that this may not be what you guys are uh uh are used to hearing when it comes

to like active inference, but hopefully
some of the ways that I'm thinking about
things, at least one or two of the
things that I've talked about kind of
informs um or helps think differently
about some of the questions that you all
u may be engaged in. Um and um yeah um I
appreciate again the
um the uh the uh the opportunity to talk
and again a lot of the uh pictures that
you've seen dur through through the
course of this talk is just the people
that um sort of um have are are critical
right to either getting the data
analyzing the data conversations that
I've had with the data or just some of
my work right and so um yeah I I would
entertain any uh questions or or
comments. Thank you again. And um it's
glad to um share some of these ideas,
some of these crazy some some a little
crazy, but it was good to share them.

Awesome.

Thank you.

All right. All right.

So, for people watching live, they can
write a question in the live chat. I'll
read one and uh we'll just go from
there. I wrote down a few things as
well. So, Hongju wrote, "How many
inevitable
overlappings could there be
between complexity versus consciousness?
Is it safe enough for us to assume
analyzing complexity is inevitably
connected to explanation of
consciousness?"

Let let me just reread it. How many

inevitable overlapping could there be
between complexity versus subjective and
phenomenal consciousness?

So maybe just how do we think about that
overlapping?

>> That's a good question.

>> I mean I'm just going to leave this uh
this uh image because it's so beautiful
and so if no one likes my answer at
least they have something good look good
to look at. So basically right like when
I think about I think that's an
excellent question because ultimately
complexity is connected to consciousness
right understanding the the different
scales of things right and then
eventually understanding how
um that can inform uh the question of
subjective experience and things like
that. Let me be a little bit more
precise with my answer.

Um you know I actually just um connected
with the Santa Fe Institute um in uh New
Mexico which they sort of pioneered
complexity science and you know one of
the things that I proposed is
understanding

uh using um the brain for example to
understand certain laws and then seeing
whether or not they scale across
different phenomenons. let's say for
example societal groups or uh you know
uh cells or uh humans or you know other
organisms right and so I think
ultimately complexity and studying it um
is important the only thing about that
is that

there have been previous work that have

shown that under complexity stability breaks down so the more complex you study something apparently right the more instable this unstable the system becomes. And so I think the challenge there or I think the opportunity is how do we study complexity?

And you know the question of can complexity bring forth some stable uh phenomenon or some stable predictions uh that mimics kind of um um you know um biological systems for example and the idea of thought and conversations or you know u systems that are adaptive and optimal and so on. So I think yes in short complexity is is is is is one of the frontiers in understanding um consciousness and even bringing us to uh a better understanding of uh how to build conscious AIs.

Awesome. All right. Next question is from Bert. He wrote there's two questions. So we'll just do one at a time. First question. Could you explain the mechanics implied in the spring visualization?

Yeah, that's a again I whoever asked that question I will um um um encourage them to read the paper but by mechanics I assume uh that the person means the sets of equations or the um the uh the rules that that shift these springs in in a particular way. So each of these so so you what I have done is that I have I I I I think about neural fields as force fields.

So you can imagine that there is a particular force

or sorry there's a moss in the in the
field that is kind of perturbing these
springs in specific directions
and the motivation for these mosses are
very uh biological

right so you can imagine

I see

a very colorful object in my in my
visual scene

Right? That colorful object you can
think of as a moss by which the springs
will kind of move towards especially if
I want to look at that object. And so
the kinematics

just thinking about neurons as particles
right under Newtonian mechanics you'll
see that they undergo kind of this $F = ma$
right these kind of ideas where the mass
and the acceleration right they will
determine the force by which that
particular thing experiences and so
that's why

uh these uh these springs right they
that's why you know I think of them as
the Newtonian law remapping, right? They
undergo these forces and these forces
are just effectively um ways by which
these neurons are allocating their
attention dynamically in space and time
in order to kind of provide sensitivity
to particular things that you want to
attend to. And so the kinematics is
basically you know uh Newtonian force
equations and uh you know putting that
under more biological constraints right
like let I'll give another very simple
example you can imagine for example when
you you normally hear when people are

like um in very high stakes situations
they'll say I couldn't see anything it
it felt like you know it felt like I I
was undergoing tunnel vision in a sense
Right.

In that sense, I would argue that
the the the force that was being exerted
in that person's visual system was so
strong that it pulled them outside of
their elast kind of like this larger
thing, right? So if I wanted to look at
something right now, my neurons need to
kind of still it needs to attend to that
area, but it kind of needs to leave a
little bit of resources in my periphery
so that I can actually see what's in my
in the corner of my eye. And so that's
what these elastic fields are for. They
kind of control for the system from
overcommitting, right? And so it's kind
of these biological constraints along
with these kind of Newtonian uh
kinematics that I've combined to create
this kind of novel computational field
of research. And so you know maybe I
wasn't may you know again whoever that
person is they could read the paper or
connect with me on Figbox. I'll love to
uh work out the equations with them if
they like. Uh but that's kind of the
idea here. And these types of things you
can imagine are useful for like computer
vision. Uh if you want to like build a
robotic eye for example, right? Does it
undergo a retinal topic mechanics? How
do you create constraints where the
robot doesn't completely converge on a
single point but has some kind of

processing in the periphery? And again these elastic fields there is no biological evidence for it. It's something that I have kind of predicted and there are particular I'm hoping experimentalists at some point uh will kind of uh give this idea some um um some uh some tension and and and explore because I think they are uh key into understanding um active sensing and active vision specifically.

>> Cool. Yeah. Just a few comments on that and then I'll ask Bert's second question.

The idea of spring-loaded systems, either coupled oscillators or springs like uh pullback attractors for joints. It's often used in active inference like in the 20122 textbook chapter 8 has examples of spring joints. And then we can generalize that a little bit more to energy based models where there's a potential function like a pullback or a gravity function

>> as you move away. So like in this flat circle with a spring pulling laterally, you could think about the Zaxis there being it's like a bowl and then there's like more pullback as you go up the edges of the bowl. And it is the generalization of Newtonian mechanics to have the basian mechanics which allows for different energy functions to exist for a given state space. So I think it's totally on the right track and I'm sure you'll find many connections as you learn about these.

>> Yeah.

Okay,

>> that's an excellent point, Daniel. And I actually want to say one last thing about that.

>> Um the I actually use the the equilibrium term this FE

right that this equilibrium force is actually the force that relaxes the spring back. So I use hook and I that is motivated by Hook's law. So as the system gets to you can imagine if I if I pull a spring really hard it could break, right? So as the system kind of gets to its what I call the elastic limit,

the spring will then pull it back and then that that that's really basically what resets it back to its equilibrium state. So you're exactly right that that gravitate that pullback spring system is also in my model and you know um you know we you know I found that you know the law that actually governs this uh remapping obeys Newtonian it's it's akin to Newtonian's uh uh uh law of gravitation right and uh you know um um it's not exactly uh Newton's law of gravitation But it's very similar. The exponent in this case is 1.6.

So yeah, excellent.

That also reminds me of uh Lachlan Kent's work on mental gravity where he explores both the qualitative metaphors and that sort of quantitative gravitational and the different exponents associated with with pullback outside of the uh energy minima. And then okay the second question of Bert

was

>> I think there's a third question. Oh.

Oh, yes. Okay. Bert wrote, "Are LLM conversations with chat GPT a huge conversation net like source? What would be possible with such large data sets?"

>> Sorry, I didn't understand that question.

>> Um, are LLM conversations, so like human talking to a language model or just to LLMs, a huge conversation net like source? What would be possible with such large data sets?

Yeah, I mean um just to clarify conversation that is human to human conversations.

Um if I understood the question correctly, Daniel and maybe you could help me. Um so the indivi the person that asked the question has proposed two scenarios. The first is an LLM and a human having a conversation, right? Like you do with chbt and so on and then a human having a conversation. And how would such data um I guess um help or um enhance uh or affect that interaction in in those two cases.

>> Yeah. Basically as the overall availability of conversation data all the all the combinations human human

>> agent agent etc. as we as the amount becomes so vast,

>> what kinds of of analyses or what kinds of models become more possible

>> other than just better recapitulation of chat?

>> Yeah. Well, I will ask the person to uh
uh put the onus on on the person that
asked the question because u you know
that's kind of what I wanted to inspire
for us to to sort of answer that
question together, right? And that's why
the data is open source, right? But a
low hanging fruit answer would be you
would just use that data to fine-tune
these models, right? And see whether or
not you get different results, right?
Um, as far as the human to human
interaction, I'm very excited about that
because, you know, one of the reasons
why I sort of thought about this this
this stuff was what if who you married
in the future was someone that you
constantly had the same thought at the
same exact time? What if that's the way
that you found your partner? That every
time you had a conversation, that person
was being synchronized on some part of
the world. I find that deeply profound
that like perhaps this entanglement akin
to like how particles are entangled
could be a way to match people right to
find like okay is my person right the
person that I'm constantly thinking
about and we already see this actually
right in the way when you really really
are close with someone you kind of
finish their sentences or sometimes
they're like oh I was just thinking
about you so I have this strong
intuition that these things already
exist we just don't have a way to
measure that and So conversation that is
my vision of one day being able to study

these spooky actions at a distance as Einstein would say right and um you know there's actually a data set right now on on conversationnet where we have two groups having a conversation at the same exact time and so people right now can just mine that data and explore that and see whether or not you kind of pick up on these resonance right and so you know that is kind of the two uh ways that I would say uh you can sort of use that data and and deploy them in the world, right? Um and you know, like I said, as we continue to build conversation that you know uh this amount of data that we're trying to collect is going to be unprecedented. And I'll say one last thing.

Normally when we think about systems, we think that when we when when they have this sort of when their internal d like you can imagine that you have this spontaneous conversations that human beings are having that in those conversations they're not like forced.

They're not scripted. They're not pretended in a sense, right?

And in that sense, if you were to use that type of data, you're able to pick up the more natural properties of what it means to be a human being, right?

You're able to pick up the more spontaneous parts of cognition because the data set, right, is very spontaneous and isn't scripted. So maybe a lot of the conversations that have been trained on on on the LLMs are conversations that are scripted and so they don't capture

real human dynamics, right? the subtleties, the noise, the the the the the stochasticity. And so again, it offers another avenue into building, you know, new types of systems, ones that are more creative, right? Rather than probabilistic, for example, right? One that are one that invent um uh uh context rather than find them, right? That's what that that's how I would think about it in this in this very moment.

>> Nice. I agree. I think it prepares the stage for systems that are part of societies of cognitive agents rather than just like an extended monologue or a book that's univocal >> is going to be continuing on a thematic thread which is good for coherence but it's still is is very single player. Um okay last question. Um, we'll address this and then also feel free just to close with like where where are you going from here? Um, what happens next? But here's the last question.

Simone,

>> sorry to cut you sorry.

>> Are there more questions or that's the actually the last question.

>> Um, the the this is the penultimate question.

>> Okay. Yeah.

>> I don't know what that means, but I I I'll take your word for it.

>> Yes. Simona wrote, "What metric would you say we would end up using to detect or measure if an AI does become conscious? Turing tests have become

outdated. Where do you see being the next measuring bar?"

>> What's the name of the person that asked the question?

>> Simona.

>> Shimona. Oh, of course.

>> Of course.

>> Yeah. She she she she she would ask that question. I know Shimona, by the way.

Anyways,

um this is actually a question that um I was asked by um one of the leading um pioneers in AI right now. I forgot. Oh, Josh Boach uh you know he asked me that same question or at least their group asked me that same question.

I am of the opinion that to build conscious AI

first of all I don't think that any human being can build a conscious AI.

Paradoxically, what I think must happen for

uh conscious AI to occur is you build kind of the the the the uh the um the foundational pieces like the clay by which that AI system can become conscious on its own independent of the architecture of the system. So let's say for example Shimona and I wanted to build a conscious AI. What we would do is we would kind of have ideas of the building blocks of consciousness and then we would use that to build the system and the architecture and then we would leave the system to its own and then come back maybe later and say hey you know are you conscious or do you start to behave very much like let's say

for example we use the human right as a measurement right as a human being. So in a sense that's a it's a paradoxical answer right in the sense that we can't build a conscious AI because consciousness I don't think is something you can directly build it's something that emerges from a system and so we would have to create or understand the properties of consciousness right as I'm trying to do with conversation that or not just me but hopefully everyone in the audience in the community and um you know build these substrates that um that um that eventually uh can um lead to AIS that become begin to have this kind of self-organization, self-emergence, right? They begin to have an intention, right? Um and so I think I think I think that's very important and if I were to go on a limb and say what is the exact property that is needed or like two things is a self-awareness of course but some concept of death

I think that the AI must invent some concept of death which obviously is going to be based upon like compute right right like the AI must be aware that if I don't power myself in some sense I will stop functioning and I think a concept of death is foundational to an AI becoming conscious. Um because I think that a lot of the things that we humans do some parts right is driven by our sense of finality right especially in the western context. I would say from a more African metaphysics death is

probably not as high in cognition. It's more about community and village and spirituality. But I think from a from a western metaphysics point of view, death is act death is a key driving force in cognition and the ways by which cognitions kind of preserve themselves in a sense.

>> Interesting. So sort of in closing, what are your next moves? What are you looking for? And how can people learn more and and collaborate and everything?

>> Yeah. Well, my f my my my ultimate move is to become top 15 on a particular list. Um,

someone, you know, but but I would say no, that was just a joke. But I would say this that my next move is, you know, continue to work with Christopher Neil. Um, you know, exploring options uh as far as uh, you know, what I want to do next as far as science. Um I also have a company uh called Figbox right which I mentioned briefly where I'm building that and trying to uh become the Trojan horse of social media uh u in a sense of trying to facilitate and connect people through human conversations and and and and u you know that's something that that that I'm working on heavily. Um I'm also uh open to collaborators right like um and um you know I'm you know I have several grants for example one of them at the Santa Fe Institute um right now um so you know um for conversation at to scale we definitely need uh people to fund the service and things like that um I also do other things in Africa where I

support like African thinkers and entrepreneurs I'm very passionate about that I think that uh science needs um to um be more interdisciplinary um um in order to to to to tap into the the knowledge and the in the abilities of people across different spectrums um as well as gender as well. So I'm very passionate about the inclusivity of science making sure women are able to contribute uh making sure that you know uh people of different cultures and races are able to contribute. I think that in order to solve complex problems, right, an interdisciplinary approach is foundational and I don't think that science is doing a very good job of that. So that's another thing that I work on. Um and so yeah, that that's kind of like um you know um kind of the my future plans and um you know um I I really am very humbled that you've um entertained to to have me speak and uh of course I would love to build relationship with you at some point as well and so thank you. Yeah,

>> thank you. Lot of food for thought, lot of material to explore. So, please be in touch and till next time. Thanks again.

>> Thanks, D.

>> Bye.

I hope that wasn't too mathematically dense, you know, because it could be intense a little bit.

Yeah. Yeah.

Yeah.